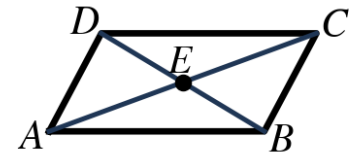


Name: _____

In $\square ABCD$ at the right, the diagonals meet at E. Use the figure for 1-5.



1. $AB = 47$. Find DC .
2. $m\angle ABC = 116^\circ$. Find $m\angle BCD$ and $m\angle ADC$.
3. $AE = 19$. Find AC .
4. $BE = 3x - 4$ and $ED = x + 12$. Find x and BD .
5. $AD = 5y + 3$ and $BC = 8y - 18$. Find y and AD .

Angles of a parallelogram.

6. Two consecutive angles of a parallelogram measure $(6x + 5)^\circ$ and $(4x + 15)^\circ$. Find x and both angles.
7. One angle of a parallelogram measures 73° . Find the other three.

Coordinates.

8. Three vertices of $\square DEFG$ are $D(-3, 2)$, $E(1, 6)$, and $G(2, -2)$. Find vertex F .
9. Check your answer to 8 using the diagonals.
10. The diagonals of $\square JKLM$ meet at $P(2, 1)$, and $J = (-3, 4)$. Find L .